

WORKSHEET - NAPLAN Year 9

Number and Algebra

Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-42664

Leo took \$72 to the 2nd hand book shop and bought a number of books.

All the books cost the same amount.

Leo paid for all the books and had no money left.

Which of these could be the amount that one book cost?

\$11 \$9 \$7 \$5
☐ ☐ ☐ ☐

2. ☒ Number and Algebra, NAP-83793

Charlie has \$20 to buy a can of soda and a chocolate bar from the shop.



How much change should Charlie be given?

\$15.70 \$16.70 \$17.10 24.30
☐ ☐ ☐ ☐

3. ☒ Number and Algebra, NAP-13227

Which expression equals a^3 ?

$\frac{a + a + a + a}{a}$ $a^2 + a$ $3 \times a$ $a \times a \times a$
☐ ☐ ☐ ☐

4. ☒ Number and Algebra, NAP-78074

Two numbers added together equal 2.

The two numbers multiplied together equal -24.

What are the two numbers?

and

5. ☒ Measurement and Geometry, NAP-66212

The diameter of a pinhead is 0.0023 mm.

Which shows this measurement written in scientific notation?

2.3×10^{-3} 2.3×10^{-2} 2.3×10^2 2.3×10^3
☐ ☐ ☐ ☐

6. ☒ Number and Algebra, NAP-36782

$36 \times \square = 20$

Which fraction makes the number sentence correct?

$\frac{3}{4}$ $\frac{4}{9}$ $\frac{5}{9}$ $\frac{2}{4}$
☐ ☐ ☐ ☐

7. ✕ Number and Algebra, NAP-95682

Lily buys 23 Gold Class tickets to the movies.

The tickets cost \$19 each.

Which of the following equations could Lily use to calculate the total cost?

- ☐ $(19 \times 20) + 3$
- ☐ $(19 \times 20) \times 3$
- ☐ $(10 \times 20) + (9 \times 3)$
- ☐ $(19 \times 20) + (19 \times 3)$

8. ✕ Number and Algebra, NAP-13288

Sharon made a number pattern using this rule:

next number = previous number $\times$  $+$ 

 and  are whole numbers.

Her number pattern is:

$-2, -1, 1, 5, 13, \dots$

What is the next number in Sharon's pattern?

- 17 19 21 29
- ☐ ☐ ☐ ☐

Worked Solutions

1. Number and Algebra, NAP-42664

$$\$72 \div 8 = \$9$$

\$9 is the only amount that can be evenly divided into \$72 with no remainder.

2. Number and Algebra, NAP-83793

$$\begin{aligned} \text{Change} &= 20 - (2.80 + 1.50) \\ &= 20 - 4.30 \\ &= \$15.70 \end{aligned}$$

3. Number and Algebra, NAP-13227

$$a \times a \times a$$

4. Number and Algebra, NAP-78074

$$\begin{aligned} 6 + -4 &= 2 \\ 6 \times -4 &= -24 \\ \therefore 6 \text{ and } -4 \end{aligned}$$

5. Measurement and Geometry, NAP-66212

$$\text{Diameter} = 2.3 \times 10^{-3}$$

6. Number and Algebra, NAP-36782

$$\begin{aligned} 36 \times \frac{5}{9} &= \frac{4 \times 9 \times 5}{9} \\ &= 4 \times 5 \\ &= 20 \end{aligned}$$

7. Number and Algebra, NAP-95682

Rewriting the equation,

$$\begin{aligned}19 \times 23 &= 19 \times (20 + 3) \\ &= (19 \times 20) + (19 \times 3)\end{aligned}$$

8. Number and Algebra, NAP-13288

Using the rule:

$$-1 = -2 \times \bigcirc + \triangle \quad \dots (1)$$

$$1 = -1 \times \bigcirc + \triangle \quad \dots (2)$$

Subtract (1) – (2),

$$-2 = -\bigcirc$$

$$\bigcirc = 2$$

Substitute $\bigcirc = 2$ into (1)

$$-1 = -4 + \triangle$$

$$\triangle = 3$$

$$\begin{aligned}\therefore \text{Next number} &= 13 \times 2 + 3 \\ &= 29\end{aligned}$$